



CS & IT ENGINEERING

Discrete Mathematics

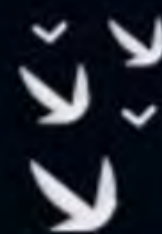
Mathematical logic

Lecture No.- 02



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Recap of Previous Lecture



Topic

Logic Part-01



Topics to be Covered



Topic

Logic Part-02



Biconditional ($\leftrightarrow$ / $\iff$)

if and only if
iff

p	q	$p \leftrightarrow q$
T	T	T
F	F	T
T	F	F
F	T	F

$$p \leftrightarrow q \equiv p \rightarrow q \wedge q \rightarrow p$$

Graph is Euler Graph if and only if degrees of all vertices are even

$$p \leftrightarrow q$$

logically equivalent

$A \equiv B$ (same behaviour)

A	B	$A \leftrightarrow B$
T	T	T
F	F	T
T	T	T
F	F	T

1) $A \equiv B$ (same behaviour)

2) $A \leftrightarrow B$ it's tautology

$|= 2$

check $(p \wedge \neg p \vee q)$ is tautology.

tautology
all are True

T
T
T
T

irrespective
of any
values of p, q
we will
always
get
whole expression True

p	q	$\neg p$	$\neg p \vee q$	$(p \wedge \neg p) \vee q$	q
T	T	F	T	T	T
T	F	F	F	F	F
F	T	T	T	T	T
F	F	T	T	T	F

Satisfiable. PW

T → tautology/valid.

	T
	T
	T
	T

↑	
↑	
↑	
↑	
	T

→ all valids are satisfiable (True)
→ all satisfiable are valid. (false)

at least
1 True

T → boys
F → girls

F → all are false

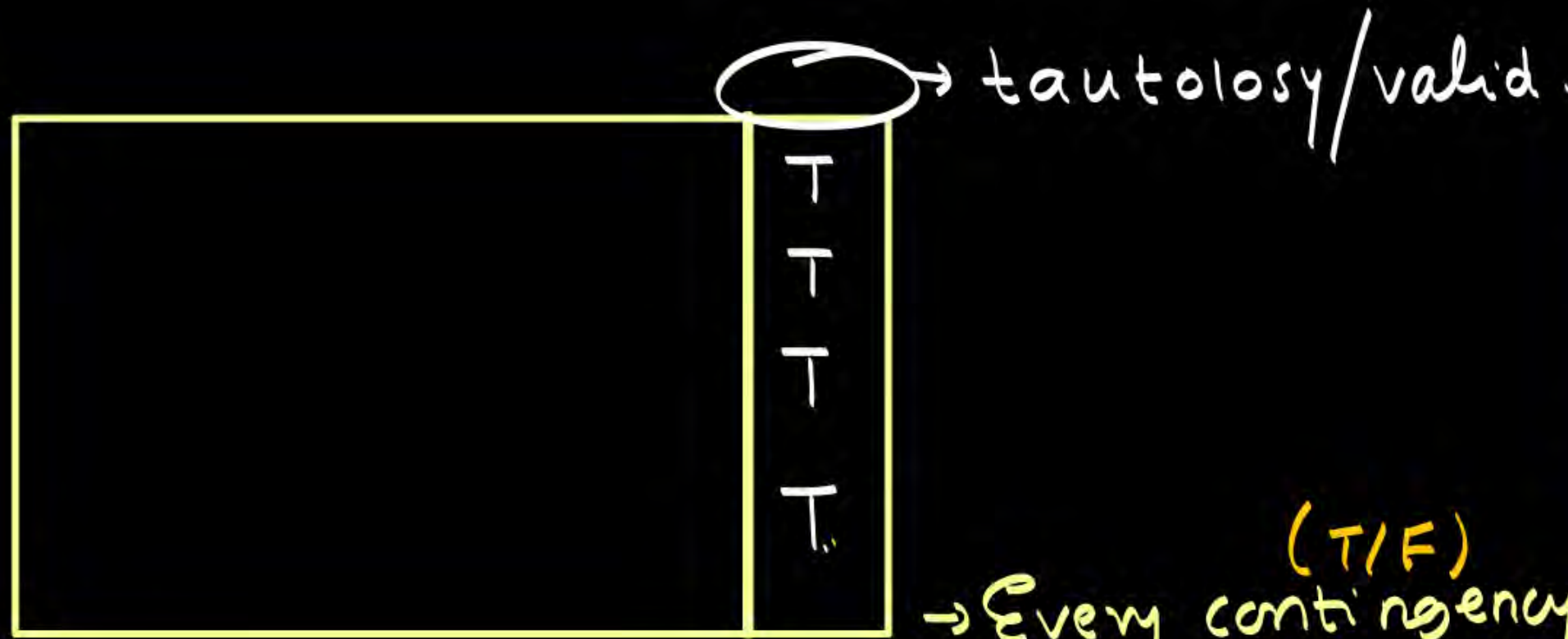
	F
	F
	F
	F

↓
Expr is
Contradiction

CONTINGENCY.

it is neither tautology nor it is contradiction.	T F
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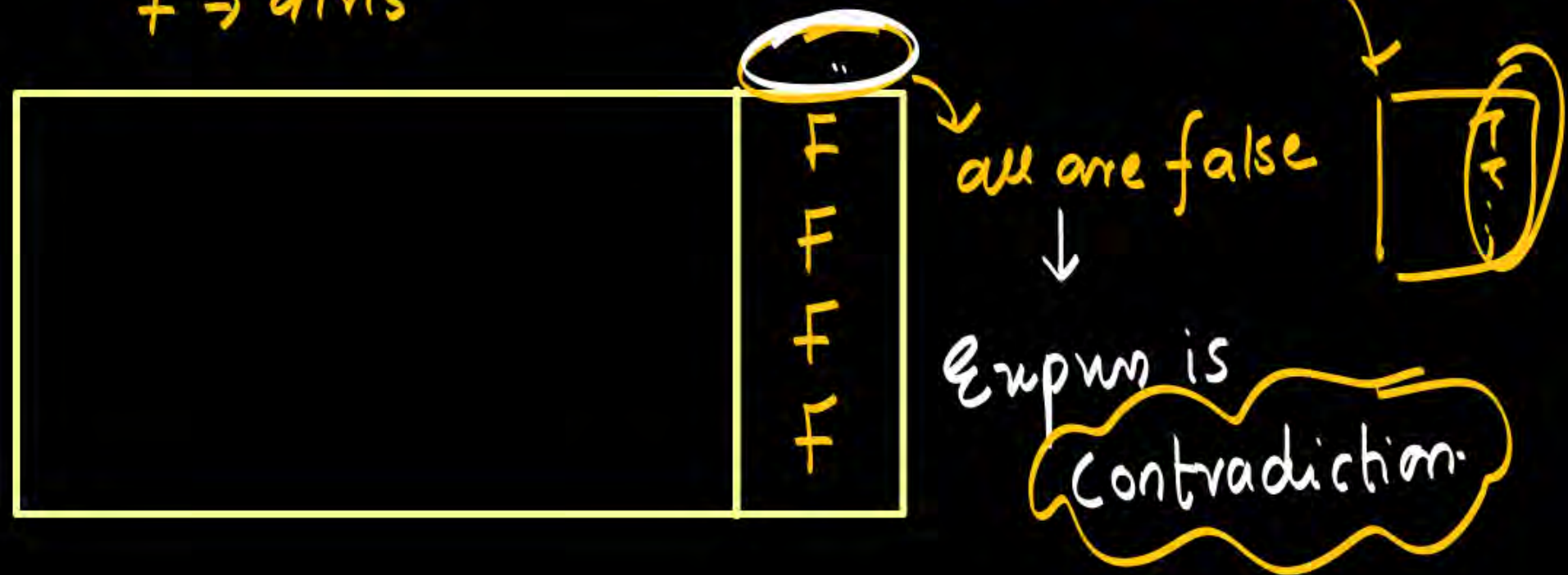
Satisfiable. PW



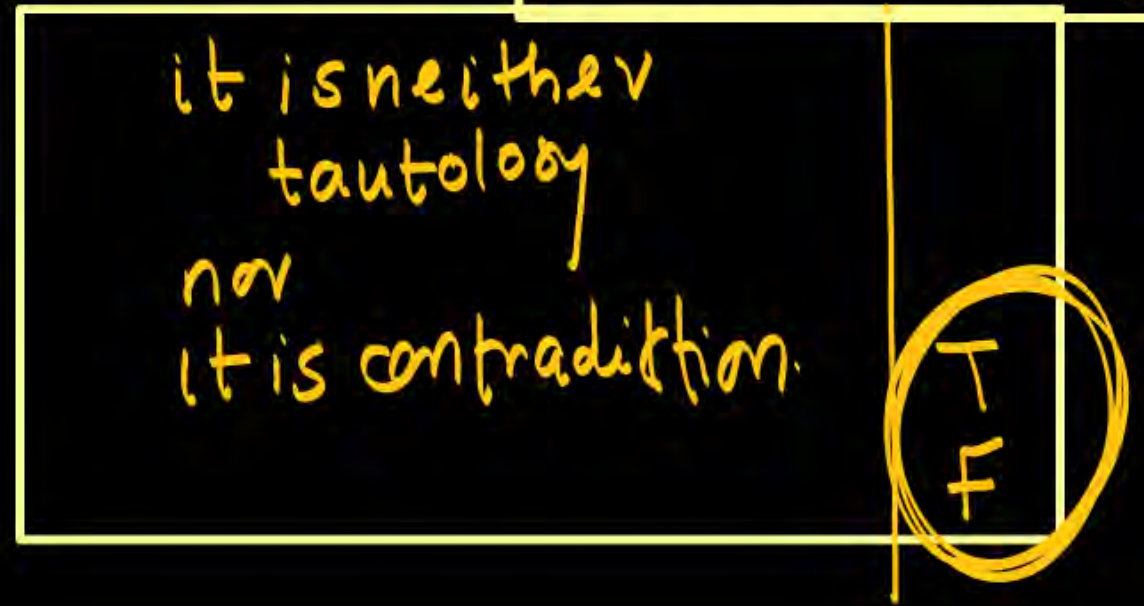
(T/F)
 → Every contingency is satisfiable (True)
 → Every satisfiable is contingency (false)

at least 1 True

T → boys
 F → girls



CONTINGENCY.



Every satisfiable is contingency (false)

	T
	T
	F

✓

	T
	F
	T
	F

✓

	T
	T
	T
	T

⊖

satisfiable but not contingency

all valids are satisfiable (True)

Every satisfiable
is contingency (false)

valid $\rightarrow$ contingency X

check its tautology?

$$1) \quad (p \wedge (p \rightarrow q)) \rightarrow q$$

tautology:

method 1: Truth table.



Type 1:

method 2: (some limitations)

$$1) \quad (\quad) \rightarrow (\quad)$$

2) tautology.

$$(p \wedge (p \rightarrow q)) \rightarrow q$$

by some means
i will try to make this
stmt as false.

Tautology $\rightarrow$ exprs will not
change.

not taut. $\rightarrow$ exprs might
change.
its truth
value.

take this false.

$$(p \wedge (p \rightarrow q)) \rightarrow q$$

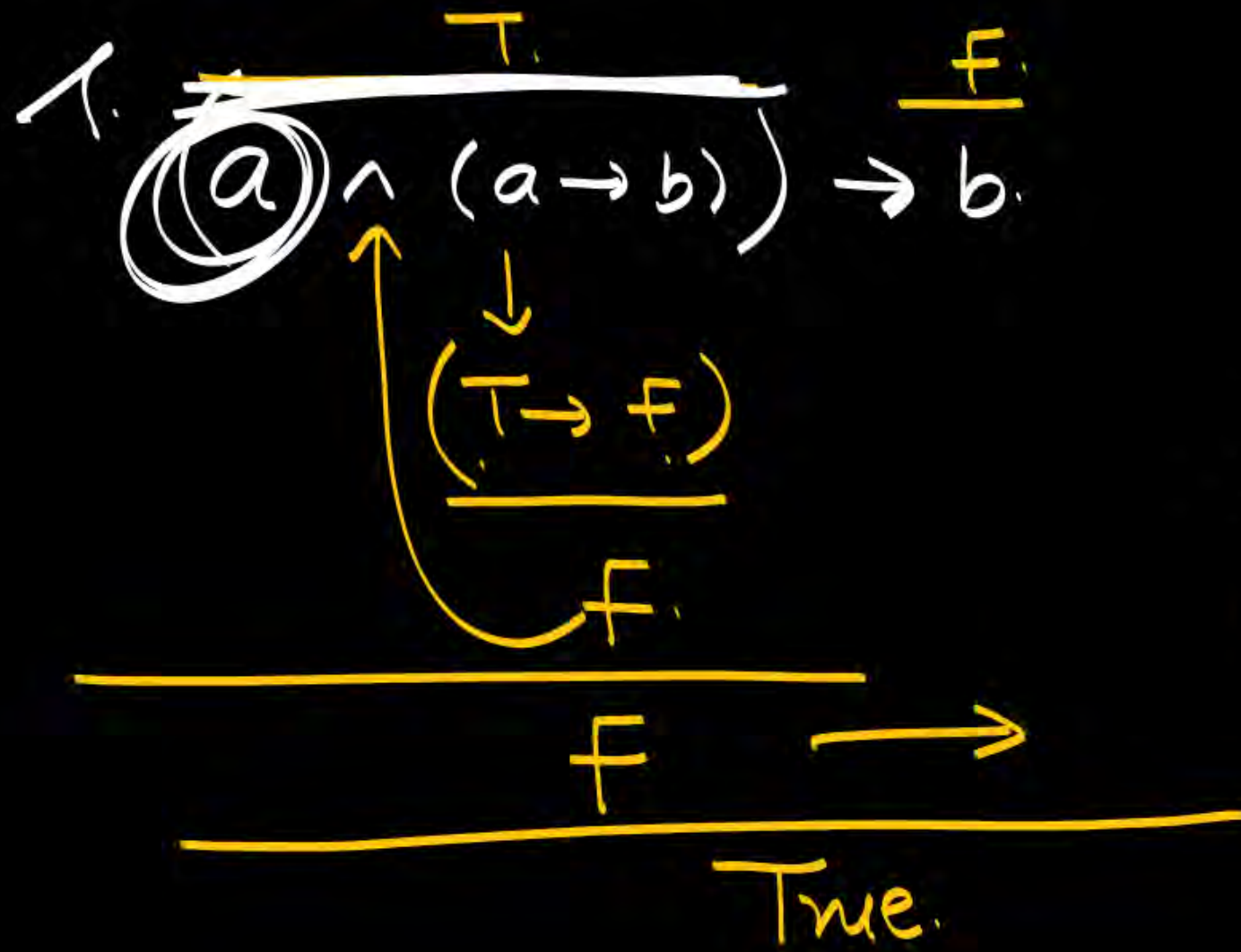
$$(T \rightarrow F)$$

$$q = F$$

$$p = T$$

F $\rightarrow$

True



Assume false

aim is to get
truth value of variable.

$$b = F$$

$$a = T$$

$$\begin{array}{c} \text{F} \quad \text{T} \quad \text{T} \quad \text{F} \\ \text{a} \wedge (\neg a \vee b) \wedge (\neg b \vee c) \rightarrow c \end{array}$$

$$\begin{array}{c} \text{F} \vee \text{F} \\ \hline \text{F} \end{array}$$

$$\begin{array}{c} \text{T} \vee \text{F} \\ \hline \text{T} \end{array}$$

$$\begin{array}{l} c = \text{F} \\ a = \text{T} \\ \neg b = \text{T} \\ b = \text{F} \end{array}$$

tautology.

F

True

$f \rightarrow _$

Invalid

$$\begin{array}{c}
 \begin{array}{c} \text{I} \\ \text{I} \end{array} \quad \begin{array}{c} \text{T} \\ \text{T} \end{array} \quad \begin{array}{c} \text{T} \\ \text{T} \end{array} \quad \begin{array}{c} \text{T} \\ \text{T} \end{array} \quad \begin{array}{c} \text{F} \\ \text{F} \end{array} \\
 \hline
 (\neg p) \wedge (p \rightarrow q) \wedge (q \rightarrow r) \rightarrow \neg r \\
 \hline
 \begin{array}{c} \text{T} \\ \text{T} \end{array} \wedge (\text{F} \rightarrow \text{T}) \wedge (\text{I} \rightarrow \text{T}) \\
 \hline
 \begin{array}{c} \text{T} \\ \text{T} \end{array} \quad \text{T} \rightarrow \text{F}
 \end{array}$$

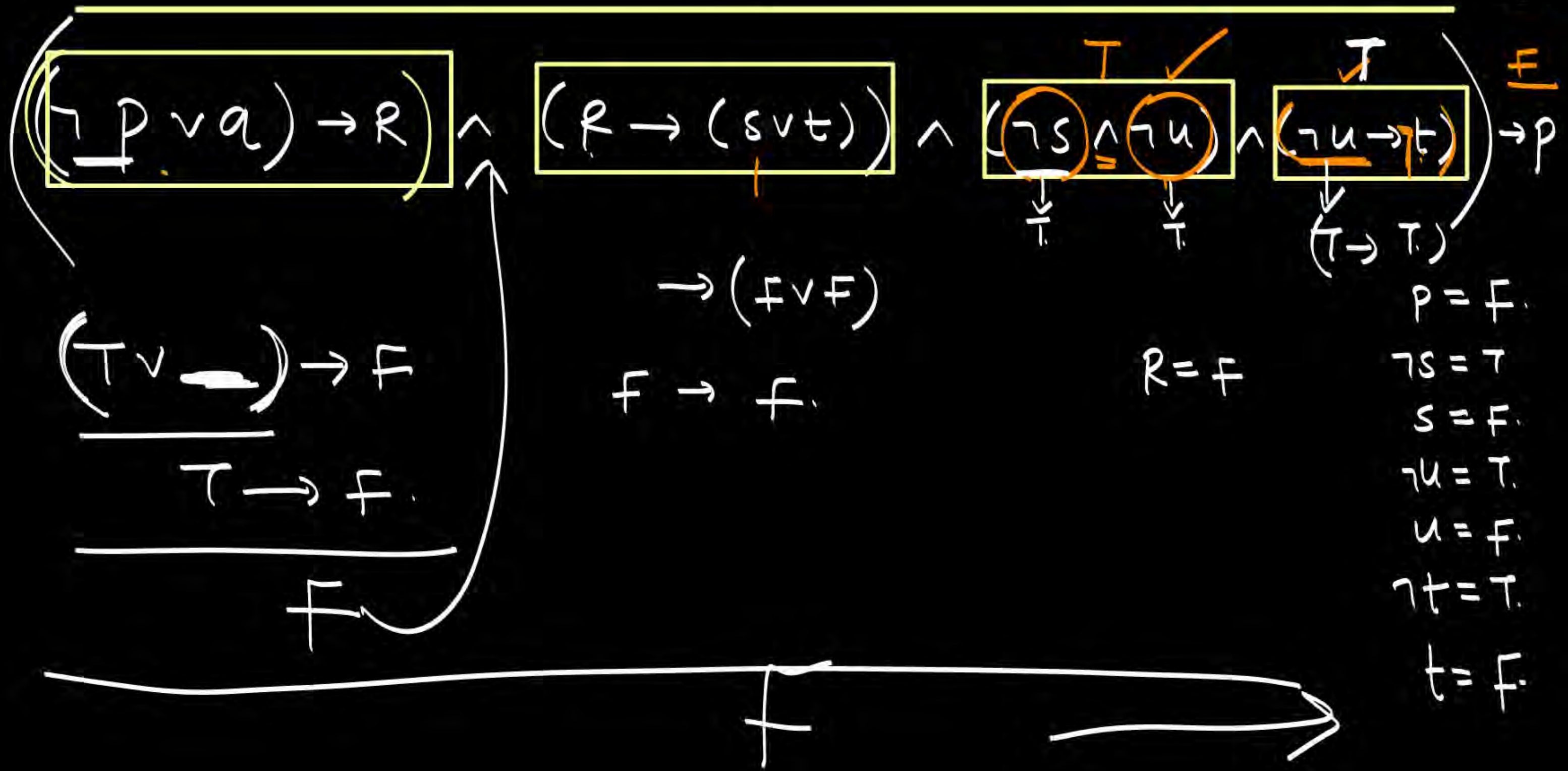
$$\neg R = F$$

$$R = T$$

$$\neg p = T$$

$$p = F$$

T



$$\begin{array}{c}
 \begin{array}{c} \text{True} \\ \hline (p \rightarrow q) \wedge (\neg r \vee s) \wedge (p \vee r) \end{array} \rightarrow \begin{array}{c} \text{False} \\ \hline (\neg q) \rightarrow s \end{array} \\
 \begin{array}{ccc} \downarrow & & \downarrow \\ (F \rightarrow F) & & (F \vee F) \end{array}
 \end{array}$$

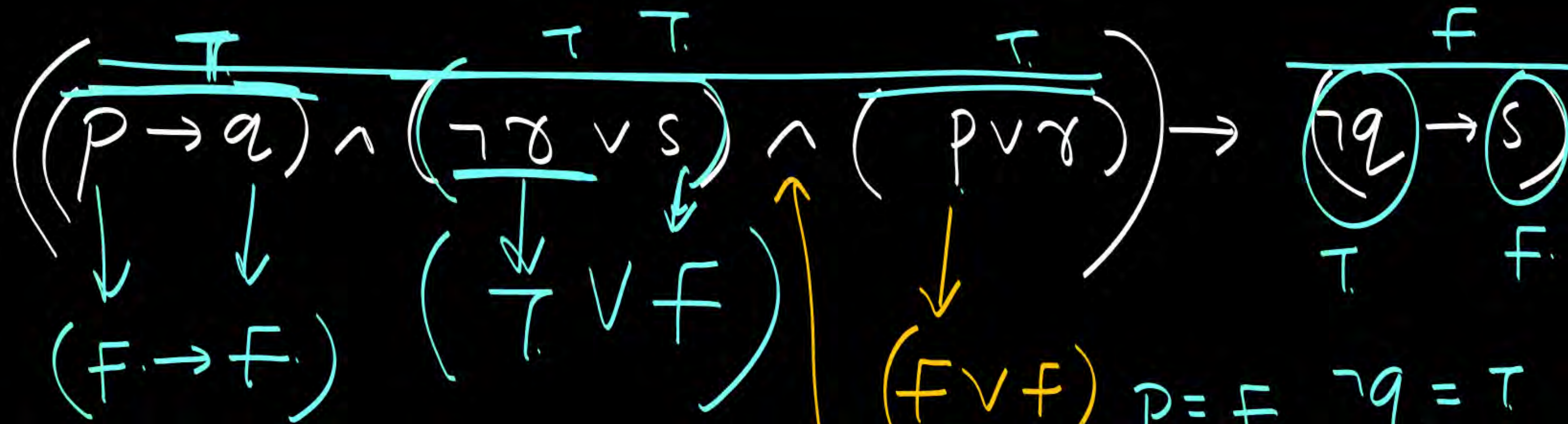
hence it is tautology.

F

True.

$$\begin{array}{l}
 \neg r = \text{True} \\
 r = \text{False}
 \end{array}$$

$$\begin{array}{l}
 p = F \quad \neg q = \text{True} \\
 q = F \\
 s = F
 \end{array}$$



hence it is tautology.

F

True.

$$[p \wedge (p \rightarrow q)] \rightarrow q$$

$$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$$

$$[(p \rightarrow q) \wedge \neg q] \rightarrow \neg p$$

$$[(p \vee q) \wedge \neg p] \rightarrow q$$

$$(\neg p \rightarrow F_0) \rightarrow p$$

$$(p \wedge q) \rightarrow p$$

$$p \rightarrow p \vee q$$

$$[(p \wedge q) \wedge [p \rightarrow (q \rightarrow r)]] \rightarrow r$$

$$[(p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow [(p \vee q) \rightarrow r]$$

$$[(p \rightarrow q) \wedge (r \rightarrow s) \wedge (p \vee r)] \rightarrow (q \vee s)$$

$$[(p \rightarrow q) \wedge (r \rightarrow s) \wedge (\neg q \vee \neg s)] \rightarrow (\neg p \vee \neg r)$$

a) $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$

b) $[(p \rightarrow q) \wedge \neg q] \rightarrow \neg p$

c) $[(p \vee q) \wedge \neg p] \rightarrow q$

d) $[(p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow [(p \vee q) \rightarrow r]$

9. For each of the following, fill in the blank with the word *converse*, *inverse*, or *contrapositive* so that the result is a true statement.

- a) The converse of the inverse of $p \rightarrow q$ is the _____ of $p \rightarrow q$.
- b) The converse of the inverse of $p \rightarrow q$ is the _____ of $q \rightarrow p$.
- c) The inverse of the converse of $p \rightarrow q$ is the _____ of $p \rightarrow q$.
- d) The inverse of the converse of $p \rightarrow q$ is the _____ of $q \rightarrow p$.
- e) The inverse of the contrapositive of $p \rightarrow q$ is the _____ of $p \rightarrow q$.



2 mins Summary

Topic

One

Logic Part-02

Topic

Two

Topic

Three

Topic

Four

Topic

Five



THANK - YOU